SIMATS ENGINEERING

ASSESSMENT TOOL 3: MCQ

COURSE NAME: COMPUTER VISION COURSE CODE:ITA0515 Slot B

COURSE OUTCOME COVERED

CO3: ***Analyze and extract image features using detection and matching techniques for effective image representation and comparison.*** **(BTL 4)**

Assessment Tool Weightage: 10%

QUESTIONS:50 Total Marks 50 Duration :60 Minutes

Annexure A

MCQ

Code of Conduct:

I R Dinesh 192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R Dinesh

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	

Application of Knowledge	**20%**	**Correctly applies concepts to problem-based questions**	**Applies concepts with minor errors**	**Limited ability to apply concepts**	**Unable to apply concepts**	
Time Management	**15%**	**Completes quiz well within time efficiently**	**Completes on time with minor delays**	**Requires extra time or rushes through questions**	**Unable to complete within given time**	
Question Interpretation	**10%**	**Interprets all questions correctly and responds appropriately**	**Misinterprets a few questions**	**Often misinterprets questions**	**Unable to understand questions**	

1. Image Normalization

An image pixel value r = 150 is normalized to the range [0,1] from 8-bit data. What is the normalized value?

Answer: C) 0.59

---

2. Image Storage Calculation

A grayscale image of size 1024 × 512 is stored using 8 bits per pixel. What is the storage requirement?

Answer: B) 512 KB

---

3. Edge Detection – Gradient Magnitude

Given Sobel gradients , , what is the gradient magnitude?

Answer: A) 10

---

4. Edge Direction Calculation

For the same gradients , , what is the edge direction?

Answer: C) 53.1°

---

5. Harris Corner Detection Response

Given det(M) = 200, trace(M) = 30, k = 0.04, what is the corner response R?

Answer: B) 164

---

6. Harris Corner Classification

If the Harris response R = +250, the pixel is classified as:

Answer: C) Corner

---

7. Hough Transform – Line Representation

A line detected at $\theta = 90°$ and $\rho = 50$ corresponds to:

Answer: B) Vertical line at $x = 50$

---

8. SIFT Scale

An image is scaled by a factor of 2. If the original keypoint scale is $\sigma = 1.6$, new scale is:

Answer: C) 3.2

---

9. PCA Dimensionality Reduction

A feature vector has 100 dimensions. PCA retains 95% variance using 20 components. What is the dimensionality after PCA?

Answer: B) 20

---

10. PCA Covariance Matrix Size

For a dataset with 50 features, what is the size of the covariance matrix?

Answer: C) 50 × 50

---

11. Image Quantization

A grayscale image is quantized from 256 levels to 32 levels. What is the number of bits per pixel after quantization?

Answer: C) 5

---

12. Image Storage Reduction

An RGB image of size 512 × 512 uses 8 bits per channel. It is converted to grayscale with 8 bits per pixel. What is the percentage reduction in storage?

Answer: C) 66.7%

---

13. Sobel Edge Response

Given Sobel kernel response values , , what is the gradient magnitude?

Answer: C) 13

---

14. Gradient Direction Calculation

For , , the gradient direction is closest to:

Answer: B) 157°

---

15. Harris Corner Detection Response

Given det(M) = 400, trace(M) = 40, k = 0.06, compute the Harris response R.

Answer: B) 304

---

16. Harris Corner Decision

For a pixel, eigenvalues of matrix M are $\lambda_1 = 120$, $\lambda_2 = 130$. This pixel is classified as:

Answer: C) Corner

---

17. Hough Transform Voting

A point contributes votes to 180 values of θ in Hough space. What is the shape of its representation?

Answer: C) A sinusoidal curve

---

18. Hough Line Detection

A detected line has parameters $\rho = 0$, $\theta = 45°$. Which line is represented?

Answer: C) Diagonal line through origin

---

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of 4×4 histograms, each with 8 bins. What is the descriptor dimensionality?

Answer: C) 128

---

20. PCA Eigenvalue Selection

Eigenvalues of covariance matrix are [40, 25, 15, 10, 5, 5]. How many principal components are needed to retain 80% variance?

Answer: B) 3

---

21. Preprocessing – Normalization Scenario

A surveillance camera outputs 8-bit grayscale images. A pixel has value 204. What is the normalized value?

Answer: C) 0.80

---

22. Image Representation Scenario

A medical image of size 2048×1024 is stored using 16 bits per pixel. What is the memory required?

Answer: B) 4 MB

---

23. Edge Detection Scenario

At a pixel location, Sobel gradients are , . What is the edge magnitude?

Answer: A) 15

---

24. Edge Orientation Scenario

Using the same gradients , , what is the edge direction?

Answer: C) 53.1°

---

25. Harris Corner Detection Scenario

For a region, det(M) = 300, trace(M) = 25, k = 0.04. What is the Harris response R?

Answer: C) 275

---

26. Harris Corner Interpretation Scenario

If the Harris response R is large and positive, the pixel corresponds to:

Answer: C) Corner

---

27. Hough Transform Scenario

A point in image space votes for all possible θ values in Hough space. What shape does this vote form?

Answer: C) A sinusoidal curve

---

28. Hough Line Detection Scenario

A detected line has parameters ρ = 40, θ = 90°. Which line is represented?

Answer: A) Vertical line x = 40

---

29. SIFT Feature Scenario

An image is scaled by a factor of 0.5. A SIFT keypoint detected at scale σ = 2.0 originally will now appear at:

Answer: A) 1.0

---

30. PCA Dimensionality Reduction Scenario

A feature vector has 200 dimensions. After PCA, 30 principal components retain 92% variance. What is the new feature dimension?

Answer: A) 30

31. Preprocessing – Bit Depth Conversion

A grayscale image originally stored in 12-bit format is converted to 8-bit. An original pixel value is 3000. What is the approximate 8-bit value?

Answer: C) 187

---

32. Image Representation – Sampling Scenario

A camera samples an image at 600 × 400 pixels. The image is down-sampled by a factor of 2 in both directions. What is the new image size?

Answer: B) 300 × 200

---

33. Edge Detection – Sobel Response

At a pixel, Sobel gradients are , . What is the edge magnitude?

Answer: C) 17

---

34. Edge Orientation Scenario

For , , the gradient direction is closest to:

Answer: C) 118°

---

35. Harris Corner Detection – Computation

Given det(M) = 500, trace(M) = 35, k = 0.05. What is the Harris response R?

Answer: A) 439

---

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents:

Answer: B) Edge

---

37. Hough Transform – Line Interpretation

A line detected with parameters $\rho = 0$, $\theta = 90°$ corresponds to:

Answer: A) Vertical line at x = 0

---

38. SIFT Descriptor Scenario

Each SIFT descriptor contains 4 × 4 blocks with 8 orientation bins each. How many values describe one keypoint?

Answer: 128

⚠️ **Note: 128 is the correct answer, but it was missing from the options you provided.**

---

39. PCA Variance Retention

Eigenvalues of covariance matrix are [50, 30, 10, 5, 5]. How many principal components are needed to retain 90% variance?

Answer: C) 3

---

40. PCA Projection Scenario

Original feature vector dimension = 120. PCA reduces it to 25 components. What is the reduction percentage?

Answer: C) 79%

---

41. Quantization Error Analysis

An 8-bit grayscale image is quantized to 64 levels. Original pixel value = 150. What is the maximum possible quantization error?

Answer: B) ±2

---

42. Image Representation Storage Comparison

Image A is 1024 × 1024 grayscale (8-bit). Image B is 1024 × 1024 RGB (8-bit per channel). What is the storage ratio (B:A)?

Answer: C) 3:1

---

43. Sobel Edge Strength Comparison

Pixel-1: . Pixel-2: . Which pixel corresponds to a stronger edge?

Answer: B) Pixel-2

---

44. Harris Corner Response Comparison

Two regions have the following Harris parameters:

Region	det(M)	trace(M)
R1	300	25
R2	350	40

Given $k = 0.04$, which region is more likely a corner?

Answer: B) R2

---

45. Harris Corner vs Edge Decision

Eigenvalues of matrix M are $\lambda_1 = 120$, $\lambda_2 = 10$. What does this pixel represent?

Answer: B) Edge

---

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at ($\rho = 60$, $\theta = 45°$). What does this intersection indicate?

Answer: C) A line in image space

---

47. Hough Transform Parameter Effect

Increasing the θ resolution from 1° to 0.5° results in:

Answer: C) Higher line detection accuracy

---

48. SIFT Scale Analysis

An object appears at scale $\sigma = 1.6$ in Image-1. In Image-2, the object is scaled by 1.5×. At what scale should the corresponding SIFT keypoint appear?

Answer: C) 2.4

---

49. PCA Variance Contribution Analysis

Eigenvalues of covariance matrix are [60, 20, 10, 5, 5]. How many principal components are needed to retain 90% variance?

Answer: C) 3

50. PCA Dimensionality Reduction Impact

Original feature dimension = 150. After PCA = 30 components. What is the percentage reduction?

Answer: C) 80%